

Understanding Mixins

This readme includes the basic of how to create new mixin.

Mixins

A mixin is a set of Grafana dashboards and Prometheus rules and alerts, packaged together in a reusable and extensible bundle. Mixins are written in jsonnet, and are typically installed and updated with jsonnet-bundler. This promotes consistency, efficiency, and maintainability across your monitoring infrastructure.

Benefits of Mixins

- **Modularity:** Break down configurations into reusable components.
- **Maintainability:** Easier to update and manage individual mixins.
- **Code Reusability:** Share common configurations across different environments.
- **Organization:** Keep your code well-structured and improve readability.

Further Reading

The monitoring.mixins.dev website has excellent further reading, as does the docs section on the [monitoring-mixins](https://github.com/monitoring-mixins) repo on GitHub. This is a really good video introduction to mixins and jsonnet.

Mixtool

Mixtool is a command-line tool designed to simplify working with mixins. It helps you:

- **Manage mixins:** Install, update, and remove mixins.
- **Generate output:** Generate the final configuration files for Grafana and Prometheus.

Setting Up Mixtool

You can install mixtool by running the script `./scripts/prepare-dev-env.sh`.

Using Mixins

Mixins offer a modular approach to building configurations for Prometheus and Grafana. Here's a basic workflow:

1. Create Configuration File (config.libsonnet): This file defines your main configuration and utilizes mixins:

```
{
  _config+: {
```

```

gitlabMetricsConfig+:: gitlabServiceMetricsConfig,
# Other configurations also go here
},

prometheusRulesGroups+:: aggregationRulesForServices(self._config),
prometheusAlertsGroups+:: alertsForServices(self._config)

}

```

- `gitlabMetricsConfig+::` containing service definition metrics configurations.
- `prometheusRulesGroups+::` Uses the `aggregationRulesForServices` to generate Prometheus rules based on the `gitlabMetricsConfig`.
- `prometheusAlertsGroups+::` Uses the `alertsForServices` mixin to generate alert rules based on the `gitlabMetricsConfig`.

2. Create Mixin File (`mixin.libsonnet`): This file combines all your individual mixins:

```

(import 'config.libsonnet') +
(import 'alerts/alerts.libsonnet') +
(import 'dashboards/dashboards.libsonnet') +
(import 'rules/rules.libsonnet')

```

3. The `generate-mixin.sh` script is designed to generate Prometheus configurations (alerts, rules, dashboards) using `mixtool` based on the provided `mixin.libsonnet` file. The outputs are stored in directories under `generated/{MIXIN_DIR}`.

```
./generate-mixin.sh {alerts|rules|dashboards|all} MIXIN_DIR
```